

Balanced quantum neural architecture search

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ARTICLE INFO

Communicated by J. Shen

Keywords:

Neural architecture search
Quantum
Evolutionary algorithm
Balance pool

ABSTRACT

In the last decade, there has been continuously increasing attention on Neural Architecture Search (NAS). The design of network architecture is aimed at automatically generating efficient neural networks in the absence of prior knowledge. Most of current oneshot NAS methods are based on weight sharing. However, there are two main problems that lead to suboptimal search results. Firstly, the large scale supernet results in a complex and difficult search for optimal subnetwork structures in the large discrete search space. Secondly, weight sharing makes the internal weight parameters of the supernet cannot always vary towards the final weights required for the optimal network structure due to the influence of different structures. These problems are responsible for the fact that the performance of the subnetworks searched by the supernet cannot represent the real performance of the subnetworks trained from scratch. To solve these problems, we propose the method based on quantum evolution and balance pool, called BQNAS, which consists of two stages. The single path supernet is trained based on weight sharing and balance pool sampling method. Then, quantum parallelism is exploited for one-hot encoding and searching for the optimal subnetwork structure based on quantum evolutionary algorithms. It can greatly improve the search efficiency of oneshot approaches in subnetwork sampling and evaluate the real performance for searching for superior subnetwork structure. Extensive experiments on two datasets show that the proposed approach outperforms the state-of-the-art ones. Specifically, BQNAS achieves a top-1 accuracy of 97.27% on CIFAR-10 and 81.36% on CIFAR-100.

1. Introduction

With the advancement of deep learning methods, Neural Architectural Search (NAS) has become an increasingly significant task. Many studies have shown architectures obtained by NAS algorithms are able to achieve higher performance compared to human-designed architectures in various real-world tasks such as image classification [1] and natural language processing [2]. Taking into account the discontinuous search process and the search space, the early methods regarded the search process as a black box optimization problem and used reinforcement learning and evolutionary algorithms to find new architectures [3,4]. However, these early methods have a huge amount of calculation costs, which limits their wide usage. Some gradient-based methods were proposed to relax the discrete search space to obtain a continuous search space [5], for example, [6] applied channel attention and partial channel connections to solve problems. However, the continuous search space means expensive computational cost.

For a single architecture, the huge search space often requires a lot of training time, so it is essential for researchers to design appropriate search strategies and performance evaluation algorithms to

find optimal architectures efficiently. Some efficient search algorithms have been proposed to lower the evaluation cost [7,8]. A large number of oneshot NAS approaches [9,10] have been proposed, which search for subnetworks by designing the search space, improving the search strategies, and selecting evaluation strategies. The simplified oneshot algorithm introduces a specific network, termed supernet, that forces the candidate subnetworks to share parameters in a complex network [11,12]. In later studies, the whole process can be divided into two steps, i.e., supernet training and candidate network searching. In weight sharing based methods, the weights stored in supernet are directly obtained to estimate the performance of focused subnetwork. However, there exists a problem that it is hard to evaluate the performance of subnetworks accurately in the space represented by the supernet. The standalone ranking is the focusing one and direct obtained ranking contains error.

A common solution to the weak correlation is to retrain the architecture with few training epochs, such as surrogate model. However, it is difficult to design the number of training epochs and evaluation algorithms. In [12], the authors employed slow-fast learning paradigm

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<https://doi.org/10.1016/j.neucom.2024.127860>

Received 28 April 2023; Received in revised form 25 January 2024; Accepted 13 May 2024

Available online 17 May 2024

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to update the architecture vectors and significantly reduced the general computation cost. Nevertheless, the fast learner largely determines the architecture, while the highly coupled slow learner makes fine-tunings. The proposed BQNAS introduces quantum computing and balance pool sampling method, in which many independent architectures can be obtained fully and efficiently in an extremely large search space.

Since Feynman proposed the first idea of quantum computing in 1982 [13], with the development of quantum computing, quantum algorithms running on quantum computers have attracted a growing interest. Quantum algorithms are increasingly proposed and proved to have great advantages over traditional algorithms such as Shor's algorithm [14,15], Grover's algorithm [16], due to their properties of superposition of quantum states and the advantage of parallel evolution. With the continuous development of machine learning and even deep learning, how to implement quantum machine learning algorithms in quantum computers or quantum systems [17] has also become the research direction of many researchers and some progress has been made such as quantum support vector machines [18], quantum perception machines [19–22], and the research related to quantum neural networks is one of them. In contrast to traditional computational analysis, where the central processing unit relies on the state of the bits, quantum computing can utilize a mass of states or superpositions of the same qubits. D. Deutsch proposed a theorem showing that deterministic quantum algorithms are much faster than any deterministic classical solution [23].

Recently, due to the parallelism and the strong population diversity and large amount of information storage of quantum system [24], Quantum Neural Networks (QNN), which combine quantum computing with neural networks, have been an increasingly popular research field [25,26]. Quantum particle swarm optimization (QPSO) has also been proposed [27], which introduces the average optimal position with fewer parameters and a much stronger global search capability than the classical particle swarm optimization [28], an optimization algorithm based on swarm intelligence. Inspired by the works, we use quantum evolution strategies to enhance the search capability of traditional evolution strategies in the vast discrete search space [29]. To ensure fairness in subnetwork selection, we propose the balance pool strategy, which alternates between supernet training and subnetworks searching to alleviate the problem of performance evaluation.

In this paper, we propose a oneshot NAS method via balance pool strategy and quantum computing, termed BQNAS, which alternates between supernet training and subnetworks searching to alleviate the problem of performance evaluation in NAS. The primary task is to encode the operation blocks in the search space, which can map traditional numerical values to quantum states. In quantum computing, the basic unit is qubit. A quantum system composed of n qubits can represent a total of 2^n states, so quantum parallel computing facilitates searching for better subnetworks. The proposed method employs quantum parallel computing to achieve low evaluation cost, and combines the advantages of different sampling methods using the balance pool strategy to find more promising population.

In conclusion, the main contribution of the proposed BQNAS are presented as following.

- (1) The quantum coding proposed by us can sufficiently reduce the search cost and can be used in quantum evolutionary algorithms.
- (2) We propose the quantum evolution algorithm to enhance the search efficiency of oneshot methods in huge search space, so that we can easily search for the optimal network architecture.
- (3) We propose the balance pool strategy to enhance the ability to find excellent subnetworks from the supernet.
- (4) Comparisons with a large number of manually designed networks and neural architecture search methods demonstrate that the proposed approach performs favorably against state-of-the-art methods.

The rest of this paper are divided into four sections. Section 2 presents a brief review to the related works. In Section 3, we give a detailed description of our proposed BQNAS. Section 4 deal with the experimental settings and results. Our conclusions are discussed in Section 5.

2. Related works

2.1. Evolutionary algorithm based neural architecture search

Neural structures have been playing an important role in the success of machine learning in recent years [30–34]. However, manually designing architectures requires a lot of labor and time. To ease the burden on scientists and reduce the influence of experience, Neural Architecture Search (NAS) has been proposed and used to focus on developing algorithms to design architectures. Evolutionary algorithm (EA) based methods evolve the architecture population according to evolution criteria [35]. They develop new promising architectures by mutating the existing architectures. In the EA-based methods [36–39], it is necessary to encode the search space to express the complete structure of the searched subnetworks. The population is continuously updating and iterating through evolutionary strategies, and a fitness evaluation score is performed on the validation set to ultimately search for the optimal subnetwork structure. To tackle the problem of candidate networks training for too long during the EA-based neural network structure search, Haoyu Zhang et al., proposed a partial weight sharing oneshot NAS framework that directly evolved complete neural network architectures [40]. Hsin-Pai Cheng et al., proposed ScaleNAS that employed oneshot learning for multi-scale (multi-path) supernet driven by grouped sampling and evolutionary search [41]. EMT-NAS was introduced by [42], which combined evolutionary multi-tasking and NAS.

2.2. Subnetwork evaluation

The shared weight strategy is used to evaluate the searched subnetwork directly after training supernet once [43]. Whereas regardless of the sampling method, biased evaluations and inaccurate subnetwork ranking scores are inevitable. FairNAS proposed that it is principal to provide equal opportunities and conditions to enhance the capabilities of all components during the training process. The fair sampling and training strategy can unleash the potential of all the modules [44]. Alternatively, excellent substructures may not be fully trained. GreedyNAS proposed a multi-path sampling method to filter out bad paths, which can reduce poorly performing operation blocks and subnetwork structures during supernet training. This made it easier to obtain potentially excellent network structures, and the evaluation performance can represent their true performance more accurately [45]. However, greedy selection may not discover enough excellent structures.

2.3. Quantum system

In classical computers, information is processed using bits, which can exist in one of two states: 0 or 1. In contrast, quantum computers use qubits (quantum bits), which can exist in multiple states simultaneously due to superposition. This property enables quantum computers to perform massive calculations in parallel. Quantum parallelism becomes particularly powerful when multiple qubits are entangled, meaning the state of one qubit is dependent on the state of another, even if they are physically separated. This ability to explore multiple possibilities in parallel is a key advantage of quantum computing and forms the basis for algorithms like Shor's algorithm for factorization and Grover's algorithm for search problems.

Chen et al. proposed quantum probabilistic neural networks (QPNN) that utilized quantum parallelism to track all possible network states, retrieve the most relevant data, and even generate new data. They

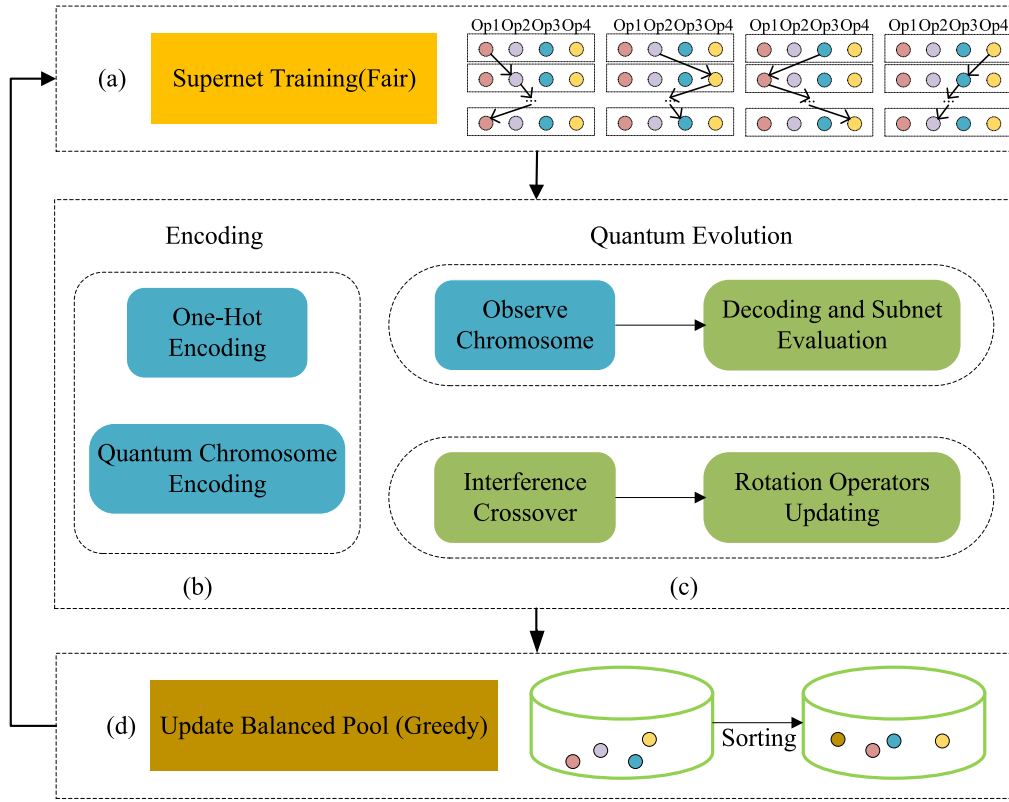


Fig. 1. The overall architecture diagram of our proposed BQNAS. (a) We build and train the supernet using weight sharing strategy. (b) Each layer requires 4 qubits to express the candidate operations. For the entire supernet with 21 candidate layers, 84 qubits are required to construct the quantum chromosome of the individual. (c) We initialize the quantum population, obtain the fitness scores of each individual in the iterative loop, and then introduce quantum global interference crossover operation in the population update stage to update the population based on the quantum rotation gates. (d) Alternate between training the supernet and evolving the subnetworks to find the final subnetwork.

demonstrated that quantum effects are useful in real classification tasks. However, this work does not consider a real quantum learning algorithm for QPNN to handle all possible layer states based on some quantum effects [46]. Patterson et al. have shown that QNN can be trained to distinguish states on noisy devices. They have further shown that gradient descent algorithms are feasible on noisy quantum devices, and the ideas of quantum computing are perfectly suited for neural architecture search [47].

As for QPSO, QICA has been proposed to deal with the problem of global optimization problem by applying quantum theory to make the algorithm more statistically robust and reasonable [48]. In [49], a new algorithm called a quantum evolutionary clustering algorithm based on watershed (QWC) was proposed. They used quantum inspired evolutionary algorithm to search the optimal clustering center and locate the edge position efficiently. [50] proposed DE-CQPSO algorithm based on the fast convergence of differential evolution algorithms and the particle diversity of crossover operators of genetic algorithms. In [51], they utilized QPSO to predict the optimal required number of edge nodes for classification model training.

3. Methodology

In this section, we propose the Balanced Quantum Evolution Neural Architecture Search (BQNAS) to address the problem of finding suitable subnetworks from supernet. The overall framework is shown in Fig. 1 and the detailed steps are outlined in Algorithm 1. Below we provide a detailed description of our proposed method.

3.1. Search space

The candidate operation blocks are based on the basis of MobileNetV2 [52]. In the addition of channel attention module, i.e.,

Squeeze-and-Excitation (SE) module and feature reconstruction (FR) module to enhance representational capacity [53,54]. The search space is shown in Table 1. As to the operation block type, MB stands for the basic block of MobileNetV2, and the first number after MB represents the expansion factor of the block. K stands for size of the depthwise separable convolution kernel. The first number after K represents the kernel size. SE and FR indicate whether to use the SE module or FR module, respectively. For example, MB3_K5_FR represents selecting the block with an expansion factor of 3, depthwise separable convolution kernel with size 5, and using the FR module. ID represents an identity mapping block with no parameters. The search space consists of 16 candidate blocks, and the total number of sub-networks in the supernet to be searched is 2^{84} , as the supernet contains 21 layers.

3.2. Quantum coding and evolution

The quantum evolutionary algorithm (QEA) is inspired by the representation of quantum states in quantum computing, where the fundamental unit of information is qubit. Each qubit $|\psi\rangle$ is in a superposition of states 0 and 1 before the observation, which can be denoted as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (1)$$

$$|\alpha|^2 + |\beta|^2 = 1$$

where α and β are complex numbers representing the distribution of the current superposition state. $|\alpha|^2$ and $|\beta|^2$ correspond to the probabilities of observing the qubit collapsing to the states 0 and 1, respectively.

Considering the extremely large discrete search space employed, we argue that traditional EA have difficulty searching for optimal subnetworks. We exploit the quantum chromosome probability amplitude superposition state in quantum evolution representation, the

Table 1
Search space of the supernet.

Index	Operation	Expansion factor	Kernel size	SE module [53]	FR module [54]	Quantum state
1	MB1_K3	1	3	–	–	0000⟩
2	ID	–	–	–	–	0001⟩
3	MB3_K3	3	3	–	–	0010⟩
4	MB3_K3_SE	3	3	✓	–	0011⟩
5	MB3_K3_FR	3	3	–	✓	0100⟩
6	MB3_K5	3	5	–	–	0101⟩
7	MB3_K5_SE	3	5	✓	–	0110⟩
8	MB3_K5_FR	3	5	–	✓	0111⟩
9	MB6_K3	6	3	–	–	1000⟩
10	MB6_K3_SE	6	3	✓	–	1001⟩
11	MB6_K3_FR	6	3	–	✓	1010⟩
12	MB6_K5	6	5	–	–	1011⟩
13	MB6_K5_SE	6	5	✓	–	1100⟩
14	MB6_K5_FR	6	5	–	✓	1101⟩
15	MB1_K3_SE	1	3	✓	–	1110⟩
16	MB1_K3_FR	1	3	–	✓	1111⟩

Algorithm 1 Framework of the proposed BQNAS

Input: Supernet M , pretraining dataset D_p , pretraining epochs N_p , fine-tuning epochs N_f , search space S , quantum population Q , interference crossover conditions C , rotation gate O , balance pool evolutionary iteration N_{bi} , maximum number of evolutionary generations G , the number of subnetworks B , balance epoch N_b ;

Output: Evaluation Network B_s .

```

1: /* Stage 1: Supernet training */
2: for  $i = 1, \dots, N_p$  do
3:   for data  $X$ , label  $Y$  in  $D_p$  do
4:     Fairly sample  $B$  single-path subnetworks from the supernet  $M$ ;

5:   if  $i \bmod N_b = 0$  then
6:     Updating balance pool  $P$  using candidate subnetworks for evolutionary  $N_{bi}$  iterations;
7:   end if
8: end for
9: end for
10: /* Stage 2: Quantum Evolutionary Algorithm */
11: for  $j = 1, \dots, N_f$  do
12:   Quantitative encoding  $S$ ;
13:   Initialize and observe  $Q$ ;
14:   Decoding and obtaining the subnetwork  $R_s$ ;
15:   if  $C$  is met then
16:     Interference crossover updating  $Q$ ;
17:   end if
18:   Updating  $Q$  with  $O$ 
19:   Updating  $B_s$  with  $R_s$  and  $Q$ 
20: end for

```

properties of measurement collapse, and the high diversity of populations with large information storage for searching the best performing subnetwork.

In our supernet, each layer requires 4 qubits to represent 16 candidate operations, as shown in Table 1. For the 21 candidate layers, a total of 84 qubits are needed to form the quantum chromosome of the current individual. For a single quantum individual, every 4 qubits represent the selection of a candidate operation block, repeated 21 times for the 21 candidate layers in the supernet. Finally, the quantum chromosome can represent the entire network structure, because the chromosome is composed of multiple qubit probability amplitudes and is in a superposition state before observation. When an observation operation is performed, the quantum chromosome collapses to a fixed state, and the corresponding network structure can be obtained through decoding.

3.3. Balance pool

We design the balance pool structure to store the more promising subnetwork structures obtained through the alternate training of the supernet and quantum evolutionary algorithm, and continuously save them until the end of the supernet training, as shown in Fig. 2. The detailed process can be described as follows:

After constructing the supernet based on the search space, all candidate operation blocks are trained with a strict and fair sampling method. Then training for N_{bp} rounds, the training process is paused, and the quantum evolutionary algorithm is used to search for the subnetwork structure of the current supernet. After N_{bf} iterations of the quantum evolution process, we sort the performances of the subnetwork structures searched in the current population, and the top five subnetwork structures in performance are added to the balance pool. In this step, we update the balance pool based on the greedy selection principle. When multiple rounds of alternate training of the super network and subnetwork evaluation, the training of the supernet finishes. In the quantum evolutionary algorithm stage, the subnetwork structures continuously updating in the balance pool are potential excellent network structures with higher probabilities. Therefore, the quantum population is initialized using the subnetwork structures in the balance pool, and then quantum evolutionary algorithm with j rounds is used to search for the optimal subnetwork structure.

Evaluation performance of the subnetwork structures directly inherited through weight sharing is significantly lower in the early stage of the supernet training because of the small number of training epochs. Despite all that, it does not necessarily mean that the subnetwork structures with lower evaluation performance are truly inferior. Therefore, the update of the balance pool is focused on subnetwork structures that have been evaluated with inferior performance in the later stage of supernet training, which largely represents their potential inferiority, so that such structures are suppressed and discarded using a greedy selection tactic. At the end of the supernet training stage, the structures in the balance pool represent potential high-performing subnetworks in various processes of the training stage, which are used as the basis for initializing the quantum population for large scale network structure search to find the subnetwork structure with the optimal performance.

3.4. Quantum evolutionary algorithm

Initialize the quantum population. This process includes constructing quantum chromosomes and preparing the equal probability quantum state. The equal probability quantum state preparation operation of the quantum chromosome, as shown in Eq. (2), is obtained by taking the tensor product of the Hadamard gate and the initial state

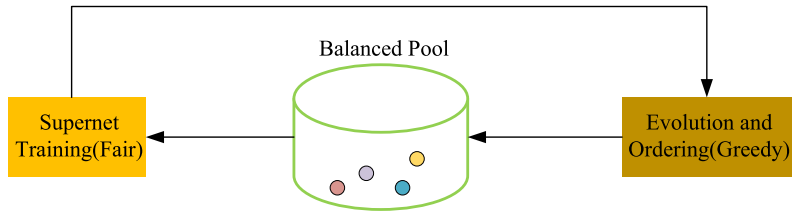


Fig. 2. Schematic diagram of the approach based on balance pool strategy.

with all $|0\rangle$ qubits. This operation is a crucial step in the initialization of the quantum population.

$$H \cdot \phi = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (2)$$

Iteration and evaluation. Observe all individuals in the quantum population at the current iteration, and collapse the quantum chromosomes in superposition to a definite state. The quantum chromosomes with genes in definite states can be decoded into specific operations for each candidate operation block in the supernet, forming a subnetwork with a determined structure. The weight sharing strategy is employed to directly use the weight parameters saved for each candidate operation block during supernet training.

$$F(I) = \phi \cdot ACC(I) + (1 - \phi) \cdot \frac{1}{P(I)} \quad (3)$$

As the primary goal of neural architecture search is to discover models that are small in size yet high in accuracy, this chapter sets the fitness function in the population to be calculated jointly from model accuracy and size. The specific calculation of fitness is described in Eq. (3). where $F(I)$ denotes the fitness value of the current individual I , ϕ is weight coefficient with a range of $[0, 1]$, $ACC(I)$ represents the model accuracy of the network structure represented by the current individual, $P(I)$ represents the number of model parameters of that network structure.

Quantum population updating process. To avoid falling into local optima and increase the expression ability and diversity of the quantum population, we introduce the quantum fully coherent interference crossover operation. This is a special crossover method for quantum chromosomes, inspired by the coherent properties of quantum systems. All chromosomes in the population participate in the crossover to the maximum extent, recombining the membership and position of each gene locus on the quantum chromosomes. If the current iteration satisfies the conditions for fully coherent interference crossover, this operation is executed. Otherwise, enter the quantum rotation update process directly.

Quantum rotation update. In the population, the update of quantum chromosomes is achieved through quantum rotation operations. In quantum systems, qubits are commonly represented by the Bloch Sphere, as depicted in Fig. 3. The transformation of a qubit's state is essentially a quantum rotation operation on the Bloch Sphere. Therefore, when performing a rotation operation on a qubit, it is determined by the state of the qubit as represented on the Bloch Sphere. Each point on the Bloch Sphere corresponds to a unique state of the qubit, where the state is determined by the angles θ with respect to the x -axis (or y -axis) and φ with respect to the z -axis.

The quantum rotation operation uses quantum rotation gates to rotate each gene position of the quantum chromosome and change the probability amplitude on the quantum gene position, thus changing the probability of collapsing into a specific state during quantum observation. We use the quantum rotation gate shown in Eq. (4) to update the quantum chromosome.

$$\begin{pmatrix} \alpha'_i \\ \beta'_i \end{pmatrix} = \begin{pmatrix} \cos(\theta_i) & -\sin(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i) \end{pmatrix} \begin{pmatrix} \alpha_i \\ \beta_i \end{pmatrix} \quad (4)$$

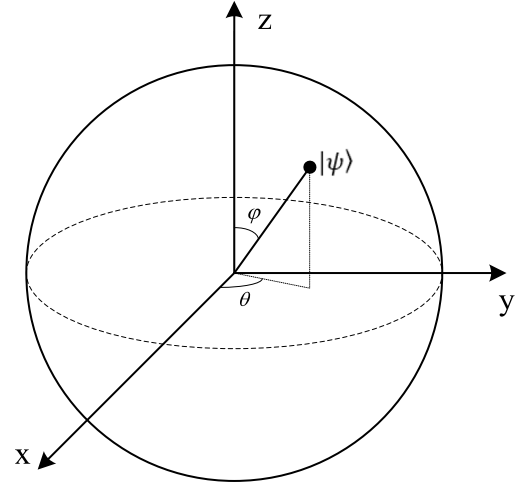


Fig. 3. The diagram of quantum state Bloch Sphere.

Table 2

Parameters used in quantum rotation gate.

x_i	b_i	$F \geq F_{best}$	$\Delta\theta_i$	$s(\alpha_i\beta_i)$			
				$\alpha_i\beta_i > 0$	$\alpha_i\beta_i < 0$	$\alpha_i = 0$	$\beta_i = 0$
0	0	×	0	0	0	0	0
0	0	✓	0	0	0	0	0
0	1	×	0	0	0	0	0
0	1	✓	0.03π	-1	+1	± 1	0
1	0	×	0.03π	-1	+1	± 1	0
1	0	✓	0.03π	+1	-1	0	± 1
1	1	×	0.03π	+1	-1	0	± 1
1	1	✓	0.03π	+1	-1	0	± 1

where $\begin{pmatrix} \alpha_i \\ \beta_i \end{pmatrix}$ represents the probability amplitude corresponding to the i th quantum bit in the current quantum chromosome, $\begin{pmatrix} \alpha'_i \\ \beta'_i \end{pmatrix}$ represents the probability amplitude obtained by the quantum rotation gate operation, θ_i represents rotation parameter corresponding to the i th qubit.

In the quantum rotation operation, the size and direction of the parameters in the quantum rotation gate depend on the relationship between the current individual and the optimal individual, as well as the value of the probability amplitude in the current quantum bit. The quantum rotation gate parameters used in deep neural network architecture search is shown in Table 2.

Where F is the current fitness value, F_{best} is the optimal fitness value, x_i and b_i correspond to the quantum states of the i th gene locus in the current solution and the i th gene locus in the optimal solution, respectively. $s(\alpha_i\beta_i)$ represents the rotation direction, $\Delta\theta_i$ represents the variation in the parameter.

Besides, in order to maintain the current best individual to update population while ensuring diversity, BQNAS adopts an elite preservation strategy by saving the individuals with the best fitness value in the current iteration and directly adding them to the next generation.

of the population. The elite preservation strategy ensures that the final best individual and its corresponding network structure obtained through quantum evolution are the global optimal solutions of the entire evolution stage.

When the maximum number of iterations is reached, the iterative process of quantum evolution ends. The quantum individual with the maximum fitness value of the observed deterministic state in the current population is the final best individual. The corresponding network structure is decoded from the best individual. We train the structure from scratch without weight sharing strategy and obtain performance metrics such as model accuracy.

4. Experimental

In this section, the experimental settings for evaluating the proposed BQNAS are presented. Section 4.1 gives a detailed description of the used image classification datasets. The implementation details of the proposed BQNAS in Section 4.2. For different networks, we give the comparisons in Section 4.3. In Section 4.4, we conduct the ablation study of the proposed BQNAS.

4.1. DataSets

For experimental evaluation, we conduct experiments on two popular image classification datasets: CIFAR-10, CIFAR-100. Both of the CIFAR-10 and CIFAR-100 datasets have 50K training RGB images and 10K testing RGB images with a fixed spatial resolution of 32×32 .

4.2. Implementation details and evaluation metrics

In the supernet training stage, the balance pool strategy is used to randomly select a candidate operation block for each mini batch to construct a single path network for training. The optimizer is SGD. Learning rate is set to 0.045 and updates using cosine annealing. The total number of training epochs is set to 20.

In the quantum evolutionary algorithm stage. In detail, the population size is set to 20, the probability of triggering full interference crossover C is set to 0.05, the fitness calculation weight is set to 0.9, and the epochs for evolution is 500.

In the alternating process of supernet training and subnetworks searching with a small number of training epochs, the number of epochs for supernet training N_b is set to 50, the number of iterations for subnetwork evaluation N_{bi} is set to 100. The size of the balance pool is set to 15.

4.3. Comparisons with state-of-the-art models

According to the parameter settings and the search space, the experimental results of our searched optimal neural network on CIFAR-10, CIFAR-100 and the current mainstream artificially designed deep neural network and NAS approaches are compared as shown in Tables 3 and 4.

In Table 3, the network searched by our proposed method achieves 97.27% accuracy on the CIFAR-10 dataset with only 2.75M parameters. The proposed method provides fair competition as well as the ability to retain potentially excellent networks, and combines the population diversity advantage of the quantum evolution strategy to search for excellent network structures with higher model accuracy but fewer model parameters in a large search space. The proposed method has greater advantages in model accuracy and model parameters compared to the manually designed network structures. In particular, our search space is based on the basic operating blocks of MobileNetV2, but the network has a 2.71% improvement in model accuracy and a 0.8M reduction in parameters on the CIFAR-10 dataset. We also compare with the current mainstream NAS methods using evolutionary algorithms as the search strategy, and the proposed method achieves a certain improvement in

model accuracy compared with these methods, while the search time of this method is 0.433 GPU days, which greatly reduces the cost of network search in terms of time cost. For NAS methods with weight sharing, such as CARS, the proposed method takes a similar amount of search time. In terms of search accuracy, the proposed method improves 0.22% over CARS. As for MOEA-PS, our method achieved similar results with lower search time. NSGA-NetV2 achieved the best performance. It used pretrained supernet, and the search cost was still expensive. In comparison, our method can express more different population individuals in the same search time, and is hard to fall into local optima and find a better network structure.

As can be seen from Fig. 4, the main reason for the smaller number of parameters in the current searched network architecture is that the searched model selects more constant mapping blocks, which to some extent makes the model have fewer candidate operation blocks as well as fewer model parameters for the same accuracy.

Further relevant experiments are conducted on the CIFAR-100 dataset as well as the comparison of the results. As can be seen in Table 4, for the manually designed network architecture, the proposed method has better performance in terms of model accuracy and model parameters. For the EA-based methods, the proposed method have a greater improvement in model accuracy, and outperform them in terms of search time and model parameters. The final network architecture searched on CIFAR-100 is shown in Fig. 5. The network architecture requires more modules containing attention mechanisms to improve model accuracy due to complex dataset in the CIFAR-100 dataset, while the constant mapping module is relatively less, resulting in a certain improvement in the number of model parameters.

4.4. Ablation study

The Ranking Correlation. The correlation between the performance of the search and evaluation phase is an important factor in the stability of the searching algorithm. The Kendall τ metric is a common measurement of the correlation between two rankings. $\tau \in [-1, 1]$, where $\tau = 1$ means the two rankings are the same, and $\tau = -1$ means they are exactly opposite. In Table 5, the correlation evaluation of sub-network performance and true performance ranking is evaluated using the Kendall τ correlation coefficient. It can be seen that the proposed method based on the balance pool achieves a Kendall τ of 0.954 in the extremely large search space. The performance of the subnetworks in the supernet is highly correlated with their true performance ranking.

The effectiveness of the balance pool strategy. As shown in Tables 6 and 7, removing the balance pool lowers the model's Top1 accuracy. Due to the huge search space, the PSO, gradient-based and EA-based search algorithms are difficult to perform well. Especially for CIFAR-100, which has more data categories and more refined classifications, the balance pool can more effectively select subnetworks.

The effectiveness of Quantum Evolution Algorithm. As shown in Table 8, it can be seen that the population size in QEA has little effect and performs better than EA. The characteristics of population diversity and high expressiveness of quantum evolutionary algorithm are beneficial to search network structures.

The searched network structures are shown in Figs. 5 and 6. Compared with the search method without balance pool, network structures searched by our proposed method with more candidate operation blocks with convolutional kernel size of 3, which may due to the strict fair sampling method used in the training phase of the supernet, so that the best network structures of each phase are retained during the update of the balance pool. We notice an interesting phenomenon that the network architectures searched by the methods always tend to select the MB6 blocks corresponding to the operations associated with larger expansion coefficients in the candidate blocks in the deep layers of the network. The size of feature map is decreasing, the information in the feature map size dimension becomes more abstract. More information in the channel dimension can help the network to further extract high level features for classifying the results.

Table 3

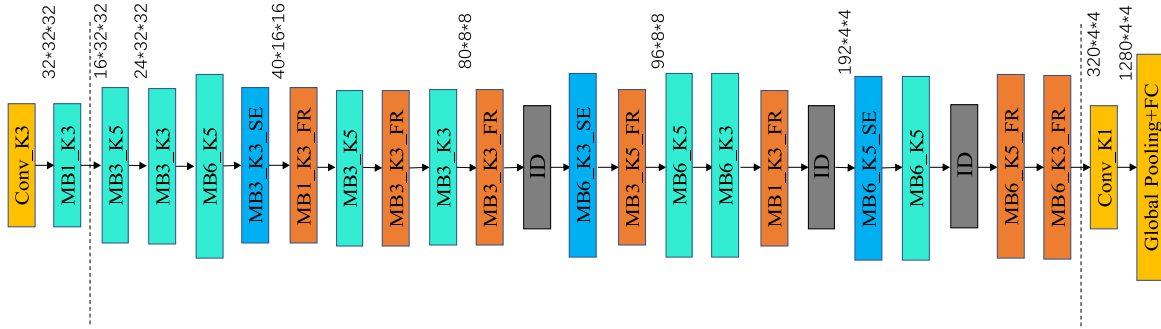
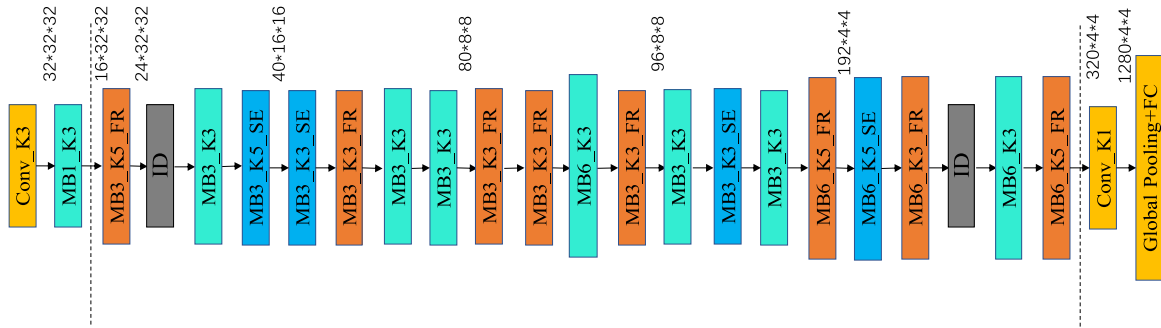
Compared with state-of-the-art network architectures on CIFAR-10. EA denotes evolutionary algorithms. QEA means Quantum evolution algorithms.

Architecture	Method	Top1 acc	Params(M)	Search cost (GPU days)
VGG [30]	Manual	93.34%	138.36	–
Resnet-101 [31]	Manual	93.57%	44.55	–
MobileNetV2 [52]	Manual	94.56%	3.51	–
DARTS [5]	Gradient	97.00%	3.3	1.5
Genetic-CNN [36]	EA	94.61%	–	17
Large-scale evolution [37]	EA	94.60%	5.4	2750
Amoebanet-A [39]	EA	96.66%	3.3	3150
CNN-GA [55]	EA	95.22%	2.9	35
NSGA-Net [38]	EA	96.15%	3.3	8
NSGA-NetV2-s [56]	EA	98.40%	6.1	2.2 ^a
CARS-D [57]	EA	97.05%	2.9	0.4
MOEA-PS [58]	EA	97.23%	3.0	2.6
Random search baseline ^b + balance pool	Random search	95.30%	2.68	0.430
Ours	QEA	97.27%	2.75	0.433

^a The search cost excludes the supernet training cost.^b Best architecture among 5 samples according to the validation accuracy after 25 epochs for supernet training.**Table 4**

Compared with state-of-the-art network architectures on CIFAR-100. EA denotes evolutionary algorithms. QEA means Quantum evolution algorithms.

Architecture	Method	Top1 acc	Params(M)	Search cost (GPU days)
VGG [30]	Manual	67.95%	138.36	–
Resnet-101 [31]	Manual	74.84%	44.55	–
MobileNetV2 [52]	Manual	77.09%	3.51	–
DARTS [5]	Gradient	82.46%	3.4	1.5
Genetic-CNN [36]	EA	74.88%	–	17
Large-scale evolution [37]	EA	77.00%	40.4	2750
Amoebanet-A [39]	EA	81.07%	3.3	3150
CNN-GA [55]	EA	77.97%	4.1	40
NSGA-Net [38]	EA	79.26%	3.3	8
MOEA-PS [58]	EA	81.03%	5.8	5.2
Random search baseline ^a + balance pool	Random search	78.00%	2.96	0.770
Ours	QEA	81.36%	2.88	0.736

^a Best architecture among 5 samples according to the validation accuracy after 25 epochs for supernet training.**Fig. 4.** Final network architecture searched on CIFAR-10.**Fig. 5.** Network architecture searched on CIFAR-100 with bp.

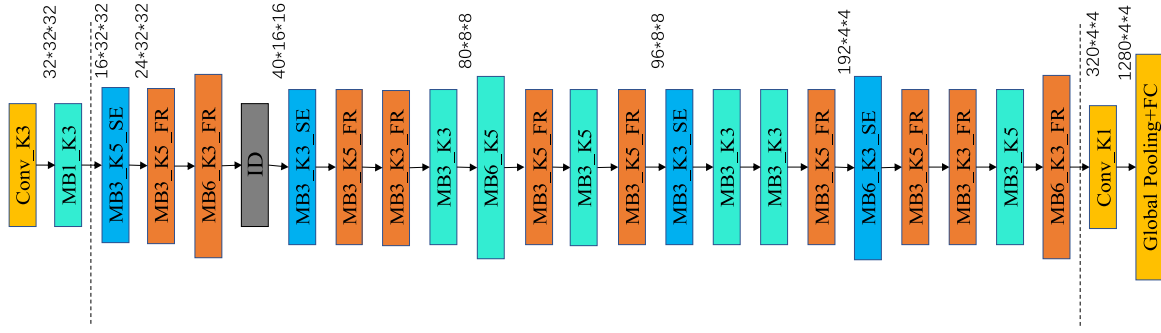


Fig. 6. Network architecture searched on CIFAR-100 without bp.

Table 5

Comparison of Kendall τ for different sampling methods.

Sampling method	Kendall τ
Random sampling	0.117
Uniform sampling ^a	0.679
Fair sampling ^b	0.918
Greedy sampling ^c	0.927
Balance pool	0.954

^a Means uniform sampling, in which the method samples randomly with equal probability in candidate architectures.

^b Means the sampling method used in FairNAS, which gives the same opportunities and conditions to improve the performance.

^c Means the sampling method used in GreedyNAS, which makes the supernet training focused on potentially superior architectures.

Table 6

Comparison of the performance and search cost of Gradient, EA and QEA with and without balance pool on CIFAR-10. BP means balance pool strategy.

Architecture	Method	Top1 acc	Search cost (GPU days)
Ours without BP	Gradient	95.33%	0.407
Ours without BP	PSO	95.28%	0.490
Ours without BP	EA	95.14%	0.429
Ours with BP	EA	95.62%	0.458
Ours without BP	QEA	96.81%	0.412
Ours with BP	QEA	97.27%	0.433

Table 7

Comparison of the performance and search cost of Gradient, EA and QEA with and without balance pool on CIFAR-100. BP means balance pool strategy.

Architecture	Method	Top1 acc	Search cost (GPU days)
Ours without BP	Gradient	80.03%	0.541
Ours without BP	PSO	80.19%	0.655
Ours without BP	EA	79.90%	0.602
Ours with BP	EA	80.65%	0.781
Ours without BP	QEA	80.83%	0.577
Ours with BP	QEA	81.36%	0.736

Table 8

Comparison of the performance in different population size with and without balance pool on CIFAR-100.

Method	Population	Top1 acc
EA	5	79.52%
EA	10	80.38%
EA	20	80.65%
QEA	5	80.01%
QEA	10	80.68%
QEA	20	81.36%

5. Conclusion

We propose an effective NAS approach named Balanced Quantum Neural Architecture Search. In the search space, we design candidate operation blocks with MobileNetV2 as the base operation block and

add attention and feature reconstruction mechanisms. In the training stage, we use the balance pool and alternate training strategy to ensure that all candidate operation blocks and subnetwork structures have the same training opportunity, while suppressing the poorly performing subnetwork structures for the training of supernet, then the population diversity and the ease of jumping out of local optima of quantum evolutionary algorithm can help us to search for diverse and excellent subnetwork structures. In the evaluation stage, a small random batch validation set is used to verify the performance of the collected networks and the corresponding individual fitness in order to reduce the time cost of the search process. The effectiveness of our proposed method is verified in the experiments as well as the comparative analysis with some current mainstream methods, and there are certain improvements and enhancements.

CRedit authorship contribution statement

Yangyang Li: Funding acquisition, Project administration, Resources, Supervision. **Guanlong Liu:** Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing. **Peixiang Zhao:** Conceptualization, Formal analysis, Investigation, Visualization. **Ronghua Shang:** Funding acquisition, Investigation, Resources, Supervision. **Licheng Jiao:** Funding acquisition, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

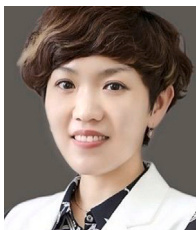
Data will be made available on request.

Acknowledgments

This work was supported in part by the Research Project of Song-Shan Laboratory under Grant YYJC052022004, in part by the National Natural Science Foundation of China under Grant 62101517 and 62176200, in part by the Natural Science Basic Research Program of Shaanxi under Grant 2022JC-45, and in part by the Fund for Foreign Scholars in University Research and Teaching Programs (the 111 Project).

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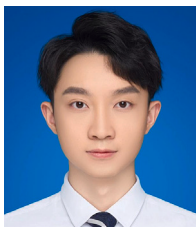
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